

Probability Theory

11.1 Introduction

In everyday life we use the word ‘probability’ or ‘chance’. For, “probably Mr. X will not attend the class today”, “it is likely to rain today”, “it is possible that I may not be able to get the handsome job.” The terms probable, likely, possible etc. are used in the same sense that is *uncertainty*. A numerical measure of uncertainty is provided by the *theory of probability*. The theory of probability was developed with the games of chance related to gambling such as tossing a coin, rolling a die, drawing playing cards from a deck and so on. These days the probability theory has become one of the basic tools in statistics.

Probability involves the study of uncertain events and uncertainty is the common factor in business and management. The demand for a product in future is uncertain because it usually varies from time to time. Tomorrow's cost will not be known until tomorrow comes. However the business decision must be made today in order to be effective tomorrow. In order to make sound decision we should try to deal with uncertainty in a systematic manner using the theory of probability. Before we introduce the probability theory, we will discuss a special counting method of *permutations and combinations* in brief, which are used in computing probabilities.

11.2 Permutations and Combinations

11.2.1 Permutations

A permutation of n different objects taken r at a time is an arrangement of r objects out of n objects with consideration given to the order of arrangement. The number of permutations of n different objects taken r at a time is given by

$${}^n P_r = n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!} \quad (1)$$

In particular when $r = n$,

$${}^n P_n = n(n-1)(n-2) \dots 2.1 \quad n = n!$$

Where $n!$ is pronounced as *factorial n*. We define $0! = 1$.

Thus, the number of permutations of three letters, a , b , and c , taken three at a time is given by

$${}^3P_3 = 3! = 3.2.1 = 6$$

The six permutations are abc , acb , bac , bca , cab and cba . Similarly, the number of permutations of three letters a , b , and c taken *two at a time* is given by

$${}^3P_2 = \frac{3!}{(3-2)!} = \frac{3.2.1}{1} = 6$$

The six permutations are ab , ba , ac , ca , bc , cb .

Note that the numbers of permutations of n objects consisting of groups of which n_1 are alike, n_2 are alike and so on is $\frac{n!}{n_1! n_2! \dots}$

Where $n = n_1 + n_2 + \dots$

Some examples of permutations are follows :

- (a) The number of ways that five differently coloured balls can be arranged in a row is

$${}^5P_5 = 5! = 5.4.3.2.1 = 120$$

- (b) The number of ways that six students can be seated on a sofa, if there are only four seats available, is

$${}^6P_4 = \frac{6!}{(6-4)!} = \frac{6.5.4.3.2.1}{2.1} = 360$$

- (c) A grocer has display space for three products. He has six products that he would like to display. The number of different arrangements of three products that he can make, is given by

$${}^6P_3 = \frac{6!}{(6-3)!} = \frac{6.5.4.3.2.1}{3.2.1} = 120$$

- (d) The number of permutations of letters in the word ‘statistics’

$$= \frac{10!}{3! 3! 1! 2! 1!} = 50400$$

11.2.2 Combinations

A combination of n different objects taken r at a time is a selection of r objects out of n objects *with no consideration given to the order* of arrangement. The number of combinations of n objects taken r at a time is given by

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!} \quad (2)$$

Note that, ${}^nC_r = {}^nC_{n-r}$

$$\text{Hence } {}^{10}C_7 = {}^{10}C_{10-7} = {}^{10}C_3 = \frac{10!}{3!7!} = \frac{10.9.8}{3.2.1} = 120$$

The number of combination of three letters a, b, c taken two at a time is given by

$${}^3C_2 = \frac{{}^3P_2}{2!} = \frac{3!}{1!2!} = \frac{3.2}{1.2} = 3$$

The combinations are ab, ac, bc . Note that ab is the same combination as ba but not the same permutation.

11.3 Basic Terminologies Used in Probability

Experiment and Events: An experiment is the activity which results in different possible outcomes or observations. If all the possible outcomes in an experiment are known beforehand and any one of the outcomes cannot be predicted then the experiment is called a *random experiment*. Performing a random experiment is called a *trial* and outcomes are termed as *events*. For example, tossing a *fair* coin is a random experiment or trial and getting a head (H) or tail (T) is an event.

Sample Space: The set of all possible outcomes of an experiment is called the sample space for the experiment. In the coin-toss experiment the sample space is: $S = \{\text{head, tail}\}$

Some experiments and their possible outcomes are given below.

Experiment	Possible outcomes
Tossing a fair coin	Head, Tail
Rolling a die	1, 2, 3, 4, 5, 6
Selecting an items from a production lot	Good, Bad
Introducing a new product	Success, Failure

Simple and Compound Events: An event is said to be *simple* if it corresponds to a single possible outcome of the trial otherwise it is *compound* or *composite* event. In rolling an *unbiased* die, the event of getting 5 is a simple event but the event of getting an even number $\{2, 4, 6\}$ is a compound event. So the compound event is the combination of two or more simple events.

Note that a simple event is also called an *elementary event* and a compound event is called *joint event*.

Equally Likely Events: Events are said to be equally likely or equally probable if each of them has an equal chance of occurring. For examples all outcomes H, T in tossing a coin or the faces numbered 1, 2, 3, 4, 5, 6 in rolling a die are equally likely because each outcome may be expected to be observed. If we draw a card from a well suffled pack, any one card may appear in the draw so that 52 different cards are equally likely.

Mutually Exclusive Events: Two events are said to be mutually exclusive if the occurrence of any one of them excludes the occurrence of the other. Events are said to be mutually exclusive if one and only one of them can take place at a time. For example, the events ‘head’ and ‘tail’ are mutually exclusive in a toss of a coin because if head occurs, tail will not occur and if tail occurs, head will not occur. Similarly the six faces 1, 2, 3, 4, 5, 6 in a roll of a die are mutually exclusive. In drawing a card from a well – suffled pack, the events a ‘red card’ and ‘a black card’ are mutually exclusive because a card cannot be red and black at the same time.

Exhaustive Events: The total number of possible outcomes of a random experiment is called exhaustive events of the experiment. In a toss of a coin exhaustive number of events are 2 i.e. {H, T}. In a roll of a die, exhaustive number of events are 6 i.e. {1, 2, 3, 4, 5, 6}. When the list of possible events that can result from an experiment includes every possible outcome, the list is said to be *collectively exhaustive* or a *sample space*.

Independent and Dependent Events: Events are said to be independent of each other if the occurrence of any one of them does not effect, is not effected by, the occurrence of any one of other otherwise they are dependent. For example if a coin is tossed twice the occurrence of head in the first toss is independent of the occurrence of head in the second toss.

In drawing cards from a pack of 52 cards, the result of the second draw will depend upon the card drawn in the first draw. However if the card drawn in the first draw is replaced before drawing the second card, then the result of second draw will be independent of the first draw.

Favourable Outcomes: The number of outcomes of a random experiment which result in the occurrence of an event are termed as outcomes favourable to the event. For example, in the tossing of a die the number of a outcomes favourable to the appearance of an even numbers are three i.e. {2, 4, 6}.

11.4 Approaches to Probability

The probability of an event is defined as a numerical value which measures the uncertainty that the event will occur i.e. the probability is *a numerical measure of likelihood of occurrence* of an event. The numerical value of the probability ranges from 0 to 1 – zero for an event which cannot occur and 1 for an event which is certain to occur. This definition does not provide the procedures how to compute the probability. Hence we will describe the three different approaches to probability and they are:

1. Classical approach (or mathematical or a priori approach)
2. Relative frequency approach (or statistical or empirical approach)
3. Subjective approach

11.4.1 Classical Approach Probability

If a random experiment results in n exhaustive, mutually exclusive and equally likely outcomes out of which m outcomes are favourable to the occurrence of an event A , then the probability of occurrence of an event (called its *success*), denoted by $P(A)$, is defined as

$$P(A) = \frac{\text{Number of outcomes favourable to } A}{\text{Total number of possible outcomes}} = \frac{m}{n} \quad (3)$$

The probability of non-occurrence of the event A (called its *failure*) is given by

$$P(A_{\bar{}}) = P(\text{not } A) = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(A)$$

$$\text{Hence } P(A) + P(A_{\bar{}}) = 1 \quad (4)$$

We can also express : $p = P(A)$, $q = P(A_{\bar{}})$ and $p + q = 1$

Note that the probability of an event lies between 0 and 1. i.e. $0 \leq P(A) \leq 1$. If $P(A) = 0$, the event A cannot occur. If $P(A) = 1$, the event A is certain to occur.

The classical probability in terms of sample space S and elementary events can be defined as:

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S} \quad (5)$$

The **limitations** of classical definition of probability are:

- (a) It fails when the total number of possible outcomes is infinite.

- (b) It cannot be used when the total number of cases are not equally likely.
- (c) It may not be possible to enumerate all the equally likely outcomes

Subjective Approach to Probability

Subjective probability may be defined as the probability assigned to an event by an individual, based on available evidences. The evidence may be just an *educated* guess or in terms of relative frequency of past happenings. Subjective probability is, in fact, the best estimate of probability in the beliefs of the individual decision maker after considering the evidence at hand. If an individual believes that it is unlikely that an event will occur, he will assign a probability close to zero to its occurrence. On the other hand if he believes that it is very likely that the event will occur, he will assign a probability close to 1. Subjective probability assessments are frequently employed when events occur only once or at most a very few times.

Subjective probability is also employed when different sources of information are in conflict. Most of the business decisions are concerned with specific, unique situations, rather than with a long series of identical situations. Thus the manager has to make use of subjective probability.

11.5 Theorems on Probability

There are two basic theorems on probability namely:

1. Addition Theorem of probability
2. Multiplication Theorem of probability

11.5.1 Addition Theorem of Probability

Case I. When events are mutually exclusive

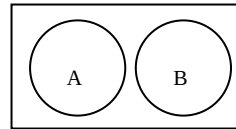
If A and B are two mutually exclusive events then the probability of either A or B occurring is the sum of the probability of individual events.

In symbols, $P(A \text{ or } B) = P(A) + P(B)$ (8)

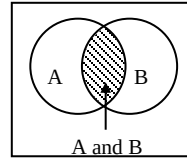
$$\text{or, } P(A \cup B) = P(A) + P(B) \quad (9)$$

The addition rule of probability is also valid for more than two mutually exclusive events. Thus we can express the probability of occurrence of one of n mutually exclusive events $A_1, A_2, \dots, A_n$ as:

$$P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_n) \\ = P(A) + P(A_2) + \dots + P(A_n)$$



Mutually exclusive events



A and B
Non-mutually exclusive events

Case II. When events are not mutually

exclusive:

If A and B are any two events then the probability of either A or B occurring is:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\text{or, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Note that if A and B are mutually exclusive $P(A \text{ and } B) = 0$ and hence Case I follows.

Exercise

Exercise - 11

- Determine the probability p for each of the following events.
 - an odd number appear in single toss of a fair die.
 - a queen appears in drawing single card from a pack of 52 cards.
 - at least one tail appears in two tosses of a fair coin;
 - a red ball appears in drawing a single ball from an urn containing 20 red and 30 green balls.
 - a tail appears in the next toss of a coin if out of 100 tosses 56 were heads.
- A die is tossed. Find the sample space of outcomes of the random experiment. Find the probability of the number appearing on the top is; (i) even (ii) less than 4 (iii) odd
- What is the probability of getting a number greater than two with an ordinary die?
- A team of scout consists of 7 person, and 4 of them are girls. A team leader is to be selected. What is probability that a boy would not be selected as a team leader? (T.U. 2061)

5. (a) A bag contains 6 white, 4 red and 10 black balls. Two balls are drawn at random. Find the probability that they will be black.

(T.U. 2072)

- (b) A box contains 4 black and 4 red balls. If 2 balls are drawn at find the probability that they are black.

- (c) In a lot of 12 pens 3 pens are defective. If two pens are drawn randomly what is the probability of getting a defective pen and a non-defective pen. *(T.U. 2065)*

6. (a) In a single throw of two dice what is the probability of getting (a) a total 8 (b) a total different from 8.

7. (a) A bag contains 30 balls numbered from 1 to 30. One ball is drawn at random. Find the probability that the number of the ball drawn will be a multiple of 5 or 7.

- (b) Twenty balls are numbered from 1 to 20. If one ball is drawn at random, what is the prob. that the ball drawn is multiple 4 or 7?

(T.U. 2058)

8. A company has 40 female employees and 60 male employees. If two employees are selected at random, what is the probability that

- (i) both will be males.
- (ii) both will be females.
- (iii) there will be one of each sex ?

9. A card is drawn from a deck of well suffled cards. What is the probability that the card will be an honor card or a 6?

10. What is the probability that in a single draw one will obtain from card-deck of 52 cards either a spade, club, heart or diamond?

11. One bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. If one is drawn from each bag, find the probability that (a) both are white (b) both are black (c) one is white and one is black.

12. (a) A can solve 90% of the problems given in a book and B can solve 70%. What is the probability that at least one of them will solve a problem selected at random?

- (b) A problem in Business Mathematics is given to five students A, B, C, D and E. Their chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{6}$. What is the probability that the problem will be solved?
- (c) The independent probabilities that the three sections of a costing department will encounter a computer error are 0.1, 0.3 and 0.3 each week respectively. Calculate the probability that there will be at least one computer error.
- (d) A person is known to hit a target in 3 out of 4 shots, whereas another person is known to hit the target in 2 out of 3 shots. Find the probability of the target being hit at all when they both try.
13. A product is assembled from three components X, Y and Z the probability of these components being defectives is respectively 0.01, 0.02 and 0.05. What is the probability that the assembled product will not be defective?
14. A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{7}$ and that of wife's selection is $\frac{1}{5}$. What is the probability that
- both of them will be selected
 - none of them will be selected
15. If a pair of dice is thrown, find the prob. that the sum is neither 7 or 11.
16. If a number is to be selected at random from integers 1 through 12, what is the probability that it is (a) even? (b) Divisible by 3? (c) Even or divisible by 3? (d) Even and divisible by 3?
17. A computer monitor is made up from four sub-assemblies: a chassis, an electronics board, a screen, and a panel of switches. Quality control investigation reveals that 1 per cent of chassis sub-assemblies, 5 per cent of electronics boards, 10 per cent of screens and 5 per cent of the panels of switches are defective. Find the probability that a computer monitor contains no defective sub-assemblies.
18. A salesman makes four calls in a day. At each call there is a probability of $\frac{1}{5}$ of making a sale. Assuming these probabilities to be independent, find the probability the salesman will make at least one sale in the day.

19. A meteorologist has forecast the probability of no rain on the Monday, Tuesday and Wednesday are 0.6, 0.5, and 0.3 respectively. Assuming that the weather from day to day is independent, what is the probability that it will rain at least once in these three days?
20. What is the probability that a leap year, selected at random, contains 53 Saturdays?
21. The probability that a contractor will get a plumbing contract is $\frac{2}{3}$ and the probability that he will not get an electric contract is $\frac{4}{9}$. If the probability of getting at least one contract is $\frac{4}{5}$, what is the probability that he will get both the contracts?
22. An MBA applies for a job in two firms X and Y. The probability of his being selected in firm X is 0.7 and being rejected at Y is 0.5. The probability of at least one of his applications being rejected is 0.6. What is the probability that he will be selected in one of the firms?
23. A fair die is rolled once. What is the probability of even score or a score divisible by 3?
24. A bank branch is asked by its head office to forward two names for transfer to a new branch. If this branch employs 12 persons – 3 males and 9 females and everybody has the same chance of being considered for transfer, what is the probability that the two persons selected would be a male and a female?